Introduction:

This is a solution and explanation for the example **“Matrix 420670”** from the series “Higher Mathematics from Maxim Yambor”.

Before going directly to the example, I would like to draw your attention to a few important points:

1. Rigor of mathematics:

This is rigorous mathematics. There are no uncertainties in this example, since they are all considered in the set of complex numbers (where only the cases 1/0 and ln(0) are uncertain), and everything was specially structured and calculated there so that the listed uncertainties did not arise. Also, the example has no incorrect entries, and the syntax is correctly written. And during the solution, it will be seen that all variables that are defined in derivatives, limits, definite integrals, and natural numbers that are defined in the sum and product of the series are finite values (except in two limits, where the variables converge to infinity, but after their solution, numbers will be obtained). The factorial is also used here, which will be taken exclusively from natural numbers.

2. The purpose of creating this example:

The purpose of creating this example is to develop mathematical intuition, which allows you to instantly see the structure of the solution.

3. Authorship:

This solution and explanation for the example **“Matrix 420670”** from the series “Higher Mathematics from Maxim Yambor” was written by the author Maxim Yambor. No artificial intelligence or any other third-party assistance was used during the creation.

4. Copyright:

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5. How to work with the example?:

For a detailed consideration of the example and explanations, I recommend stopping at each stage. This will help you better understand each stage of the solution.

After these explanations, you can proceed to the solution and explanation for the example **"Matrix 420670".**

Solution and explanation for the example "Matrix 420670"

Beginning:

Let's look carefully at the example. We see a 2 by 2 matrix, from which the determinant "det" is taken. Let's denote the matrix as "A" with elements "a11, a12, a21, a22". To find the determinant of the matrix "A" "det(A)", we need to subtract the product of the elements of the main diagonal from the product of the elements of the side diagonal: "det(A) = a11 × a22 - a12 × a21". Now that we know the algorithm, let's first calculate each element of the matrix, and then calculate the determinant itself.

Calculating the element “ a11 ”:

Let's look carefully at the element “ a11 ”. We see a limit with eight variables: "y, a, m, b, o, R, s, x". If we calculate them, we will get real numbers that lie in the interval from "8" inclusive to "229" exclusively.

Now consider the derivative with respect to "t", which is written in the numerator of the limit. We see that the derivative is taken from a certain function. If we look carefully at this function, it becomes clear that it does not depend on the variable "t", but only on the variables specified in the limit. Since the function does not contain "t", for this operation it is a constant. And the derivative of the constant is 0. Therefore, in the numerator of the limit we get 0.
Now let's look at the denominator. There is written the division of two elements:

Upper element (numerator of the fraction in the denominator):
Here we see the sum of the series, in which the upper and lower limits of the summation coincide "Σ(p=221 to f=221)". Inside is written the definite integral from "p" to "f" along dx. In this integral is written the function of the variables that are indicated in the limit. Since the limits of the summation coincide, then in this integral the limits of integration will also coincide, therefore this integral is equal to 0. But there is also written the factorial which will be taken from 0. And "0!" this is 1. And since in the sum of the series the limits of addition coincide, we simply get this number 1.
So in the upper element (the numerator of the fraction in the denominator) we got 1

The lower element (the denominator of the fraction in the denominator):
Here we see the product of the series, in which the upper and lower limits of multiplication also coincide "Π(f=42 to p=42)". The derivative of the function with respect to "t" is written in

it. If we look carefully at this function, we will see that the variable "t" is not written there, so this function does not depend on this variable. Therefore, when we take the derivative with respect to "t", this function will be like a constant. And the derivative of the constant is 0. Therefore, here we get 0 . But there is also a factorial written there that will be taken from 0. And "0!" this is 1. Therefore, it turns out simply 1. And since in the product of a series the limits of multiplication coincide, the result will be simply the number 1.
So in the lower element (fraction in the denominator) we got 1.

We have formed an expression inside the limit "0 / (1 / 1)". Since zero divided by any number is equal to 0. Therefore, under the limit sign we have 0. The limit of 0 is also equal to 0.

Conclusion: the first element of the matrix “ a11 ” is equal to 0. Now let's move on to the next element.

Calculation of the element “ a12 ”:

Now let's look carefully at the element “ a12 ”. We see a double definite integral over dxdy. There are two limits of integration written there. In the first, the upper limit of integration "42" and the lower limit of integration "42" are written, and in the second, the upper limit of integration "221" and the lower limit of integration "221". In the double definite integral there is a function written there that depends on the variables "x" and "y". That is, " ∫ ∫ (42 to 42)(221 to 221)(f(x,y))dxdy". And since the limits of integration coincide, this will be equal to 0.

Conclusion: the second element of the matrix “a12” is equal to 0. Now let's move on to the next element.

Calculation of the element “ a21 ”:

Now let’s look carefully at the element “ a21 ”. We see the sum of the series in which the upper and lower limits of the summation coincide " Σ(f=67 to p=67)". Let’s look carefully at the numerator in the sum of the series. There the division of two elements is written:

The upper element (the numerator of the fraction in the numerator):

Here we see the sum of the series "Σ(m=68 to a=134)". In the sum of the series, a number is written that when calculated will give a result that is not equal to 0. This number, as we see, is raised to a power. In the power, the derivative with respect to “t” is written where the function is written. As we see, the variable “t” is not written in the function. And this means that the function does not depend on the variable “t”. Therefore, when we take the derivative with respect to “t”, the function will be a constant. And the derivative of the constant is 0. Therefore, we have the number raised to 0. Any number except 0 raised to the power of 0 will always give 1. There, the derivative with respect to “t” is also added to 1 where the function is written. As we can see, the variable “t” is not written in the function. And this means that the function does not depend on the variable “t”. Therefore, when we take the derivative with respect to “t”, the function will be a constant. And the derivative of the constant is 0. Therefore, in the sum of the series we will have “1+0” and this is equal to 1. Since our lower limit of summation is 68 and the upper limit of summation is 134, then when we apply the sum of the series for 1, this means that we need to add “1” 67 times. And this is equal to 67. Now we see that 67 is raised to the power. In the power, the derivative with respect to t is written where the function is written. As we can see, the variable “t” is not written in the function. And this means that the function does not depend on the variable “t”. Therefore, when we take the derivative, the function will be a constant. And the derivative of the constant is 0. But there is also a factorial written there that will be taken from 0. And "0!" This is 1. So we have to the power of 1. Any number raised to the power of 1 will result in the same number. Therefore, "67^1" will be equal to 67. A definite integral is also added to this number in which the upper and lower limits of integration are 191 in dt. In this integral, a function of the variable “t” is written. Since we already know that when the limits of integration in a definite integral coincide, it will always be equal to 0. So we have "67+0" which is equal to 67.
So in the upper element (the numerator of the fraction in the numerator) we got 67.

The lower element (the denominator of the fraction in the numerator):
Here we see the sum of the series in which the upper and lower limits of summation coincide "Σ(y=19 to a=19)". It contains a limit with the variable "x" which coincides to infinity. In the limit, a root with a power is written "x". In the root we have a positive function written in which only the variable "x" is in the numerator in where it is written:
"4ln(((5+arcsin(1/xe)+11)/(2πe+arccos(a)^e))^2)". There we have "arcsin(1/xe)". When we apply the limit, then "1/xe" will be an infinitely small number that converges to 0. And "arcsin(0)" will be equal to 0. Therefore, it turns out "4ln(((5+11)/(2πe+arccos(a)^e))^2)". Therefore, this "x" in this root will not give uncertainty. So when we apply the limit, the root with the power of "x" in which the addition function is written will turn out to be 1. There, in the sum of the series, the factorial is written, which will be taken from 1. And "1!" this is 1. So in the sum of the series we got 1. And since the limits of the summation coincide, we will simply get the number 1.

So in the lower element (the denominator of the fraction in the numerator) is equal to 1.

Therefore, in the numerator, we get "67/1" which is equal to 67.

Now we move on to the denominator in the sum of the series. There is also written the division of two elements:
The upper element (the numerator of the fraction in the denominator):
Here we see the sum of the series in which the upper and lower limits of the summation coincide "Σ(m=19 to b=19)". There we see a definite integral where the upper limit of integration is “m”, the lower limit of integration is “b” by dx. This integral is a function of the variable “x”. Since in our sum of the series the limits of the summation coincide, in the definite integral the upper and lower limits will also coincide. Therefore, this integral will be equal to 0. There, in the sum of the series, the factorial is written, which will be taken from 0. And "0!" This is 1. So the sum of the series will be 1. And since the sum of the series has the same summation limits, we will simply get the number 1.
So the upper element (the numerator of the fraction in the denominator) will be 1.

The lower element (the denominator of the fraction in the denominator):
There we see the sum of the series in which the upper and lower summation limits also coincide "Σ(o=19 to R=19)". In this sum of the series, the derivative with respect to "x" is written, which is taken from the function. Now let's look carefully at this function. We see that the variable "x" is not written there. So this means that this function does not depend on the variable "x". Therefore, when we take the derivative with respect to "x", the function will be considered a constant. And the derivative of the constant is 0. There, the sum of the series also contains the factorial, which will be taken from 0. And "0!" this is 1. So in the sum of the series we get 1. And since in the sum of the series the limits of the summation coincide, we will simply get the number 1.
So the lower element (the denominator of the fraction in the denominator) will be equal to 1.
Therefore in the denominator we got "1/1" and this is 1.

Now in the sum of the series we have "67/1" which is equal to 67. And since in the sum of the series the limits of the summation coincide, we will simply get the number 67.

Conclusion: the third element of the matrix “ a21 “ is equal to 67.
Now let's move on to the next element.

Calculation of the element “ a22 “:

Now let's look closely at the element “ a22 “. We see the product of the series in which the upper and lower limits of multiplication coincide "Π(f=42 to p=42)". Now let's look closely at the numerator of the product of the series. There is written the division of two elements:

Upper element (numerator of the fraction in the numerator):
There is written a definite integral over dx in which the upper limit of integration is 84 and the lower limit of integration is 42. Now let's look closely at what is written in this integral.
There we see a 2 by 2 matrix, from which the determinant "det" is taken. Let's denote the matrix “ A' “ with elements "a'11, a'12, a'21, a'22". To find the determinant of the matrix “ A' “ "det(A')", we need to subtract the product of the main diagonal elements from the product of the side diagonal elements: "det(A') = a'11 × a'22 - a'12 × a'21". Let's calculate each element of the Matrix:
Element “ a'11 “:
There we see the derivative with respect to “t” from which the function is taken. Let's look carefully at this function. We see that the variable “t” is not written there. And this means that the function does not depend on the variable “t”. Therefore, when we take the derivative with respect to “t”, the function will be like a constant. And the derivative of the constant is 0.
Therefore, the element “ a'11 “ is equal to 0.
Element “ a'12 “:
There we see a definite integral with respect to dy in which the upper and lower limits of integration are 42. In this integral, a function with the variable “y” is written. And since the limits of integration in this integral coincide, it will be equal to 0.
Therefore, the element “ a'12 “ is equal to 0.
Element “ a'21 “:
There the sum of the series in which the lower limit of summation is 68 and the upper limit of summation is 134. That is, "Σ(a=68 to b=134). In this sum of the series, the limit is written in which the variable "c" goes to infinity. The limit is written in the root with the degree "c". In the root, we have a positive function, and if we look carefully at it, we see that the variable "c" is not written there, which means that the function does not depend on this variable. So when we apply the limit, the root with the degree "c" in which a positive function is written that does not depend on the variable "c" will be 1. So in the sum of the series, 1 is written. And since our lower limit is 68 and the upper limit is 134, then when we apply the sum of the series for 1, this means that we need to add 1 67 times. And this is equal to 67. So the element " a'21 " is equal to 67.
The element " a'22 ":
There the definite integral of dy is written where the upper limit of integration is 84 and the lower limit of integration is 42. Let's look carefully at what is written in this integral. There we see a function with the variable "y". This function is raised to a power. The derivative with

respect to "t" is written in the power, which is taken from the function. If we look carefully at this function, we will see that it does not depend on the variable "t", which means that the function does not depend on this variable. Therefore, when we take the derivative with respect to "t", the function will be like a constant. And the derivative of the constant is 0. So we have 0 written in the power. Any number other than 0 raised to the power of 0 will be equal to 1. Therefore, we have the number 1 in this sum of the series. The derivative with respect to "t", which is taken from the function, is also added to this number. If you look carefully, the variable "t" is also not written there, so this function also does not depend on this variable. And as we already know that if we take the derivative with respect to “t”, then this function will be like a constant. And the derivative of the constant is 0. Therefore, we have "1+0" which is equal to 1. So in this integral 1 is written.
When we apply this integral to 1 we get "1×y". And since this integral is definite, according to the Newton-Leibniz formula we write it like this "1×84-1×42" where instead of the variable "y" we substitute the numbers that are written in the upper and lower limits of integration (84 and 42). And this is equal to 42.
So the element “ a'22 “ is equal to 42.
Now since we have counted the elements in the matrix “ A' “ now we will count the determinants of this matrix. We will get "0×42-0×67" which is equal to 0. So the determinant of this matrix is zero.
Now we see that in this integral we got 0. But there is also written the factorial which will be taken from 0. And "0!" This is 1. So in this integral we got 1. When we apply this integral to 1 we get "1×x". And since this integral is definite, by the Newton-Leibniz formula we write like this "1×84-1×42" where instead of the variable "x" we substitute the numbers that are written in the upper and lower limits of integration (84 and 42). And this is equal to 42.
So the upper element (the numerator of the fraction in the numerator) is equal to 42.

The lower element (the denominator of the fraction in the numerator):
There we see two terms. Let's look at the first term. The derivative with respect to "y" is written there. It contains the limit where the variables "y" and "a" go to "7" and "3". In this limit a function with the variables "y" and "a" is written. When we apply this limit, then we will get a number that will not depend on these variables. Then after that we will take the derivative with respect to "y" of the number that will no longer depend on the variable "y". And this number will be like a constant. And the derivative of the constant is 0. Therefore, there we get 0. But there is also a factorial written there that will be taken from 0. And "0!" This is 1.
So in the first term there will be 1. Now let's look at the second term.

There is written the derivative with respect to "a". It contains the limit where the variables "a" and "m" go to "42" and "67". In this limit is written the function with the variables "a" and "m".

When we apply this limit, we will get a number that will not depend on these variables. Then after that we will take the derivative with respect to "a" from a number that will no longer depend on the variable "y". And this number will be like a constant. And the derivative of the constant is 0. Therefore, there it turns out 0.
So the second term will be equal to 0.
So we got "1+0" which is equal to 1.
So the lower element (the denominator of the fraction in the numerator) will be equal to 1.
So in the numerator we got "42/1" which is equal to 42.
Now we move on to the denominator in the product of the series. And the division of two elements is also written:
The upper element (the numerator of the fraction in the denominator):
There we see two terms. Let's look at the first term. The derivative with respect to "m" is written there. It is written in the limit where the variables "m" and "c" go to "67" and "3". In this limit a function with the variables "m" and "c" is written. When we apply this limit, we get a number that will not depend on these variables. Then after that we will take the derivative by "m" from a number that will no longer depend on the variable "m". And this number will be as a constant. And the derivative of the constant is 0. Therefore, it turns out 0. But there is also a factorial written there that will be taken from 0. And "0!" This is 1.
So in the first term there will be 1. Now let's look at the second term.
There is a derivative written there with respect to "b". In it is written a limit where the variables "b" and "l" go to "69" and "3". In this limit a function with the variables "b" and "l" is written. When we apply this limit, we will get a number that will not depend on these variables. Then after that we will take the derivative with respect to "b" from a number that will no longer depend on the variable "b". And this number will be as a constant. And the derivative of the constant is 0. Therefore, there it turns out 0.
So the second term will be equal to 0.
So we got "1+0" which is equal to 1.
So the upper element (the numerator of the fraction in the denominator) will be equal to 1

The lower element (the denominator of the fraction in the denominator):
There we see two terms. Let's look at the first term. The derivative with respect to "o" is written there. In it is written the limit where the variables "o" and "v" go to "191" and "3". In this limit a function with the variables "o" and "v" is written. When we apply this limit, we get a number that will not depend on these variables. Then after that we will take the derivative with respect to "o" from a number that will no longer depend on the variable "o". And this number will be like a constant. And the derivative of the constant is 0. So there it turns out 0. But there is also written the factorial which will be taken from 0. And "0!" That is 1.
So in the first term there will be 1.
Now let's look at the second term.

There is written the derivative with respect to "R". It contains the limit where the variables "R" and "l" go to "221" and "3". In this limit a function with the variables "R" and "l" is written. When we apply this limit, we get a number that will not depend on these variables. Then after that we will take the derivative with respect to "R" from a number that will no longer depend on the variable "R". And this number will be like a constant. And the derivative of the constant is 0. Therefore, it turns out 0.
So the second term will be equal to 0.
So we got "1+0" which is equal to 1.
So the lower element (the denominator of the fraction in the denominator) is equal to 1.
So in the denominator we got "1/1" which is equal to 1.
Now we have in the product of the series "42/1" and this is equal to 42. Therefore, we got 42 in the product of the series. And since we have the limits of multiplication in the product of the series coincide, we will simply get the number 42.
Conclusion: the fourth element “a22” is equal to 42. Now let's move on to the calculation of the determinant of this matrix.

Calculating the determinant of Matrix “A” "det(A)":

So we have calculated each element of Matrix “A”. Now we can substitute our results into this formula "det(A) = a11 × a22 - a12 × a21".
We get "det(A) = 0 × 42 - 0 × 67" which is 0.
So the determinant of this matrix will be 0.

Therefore, this example has the answer “0”!